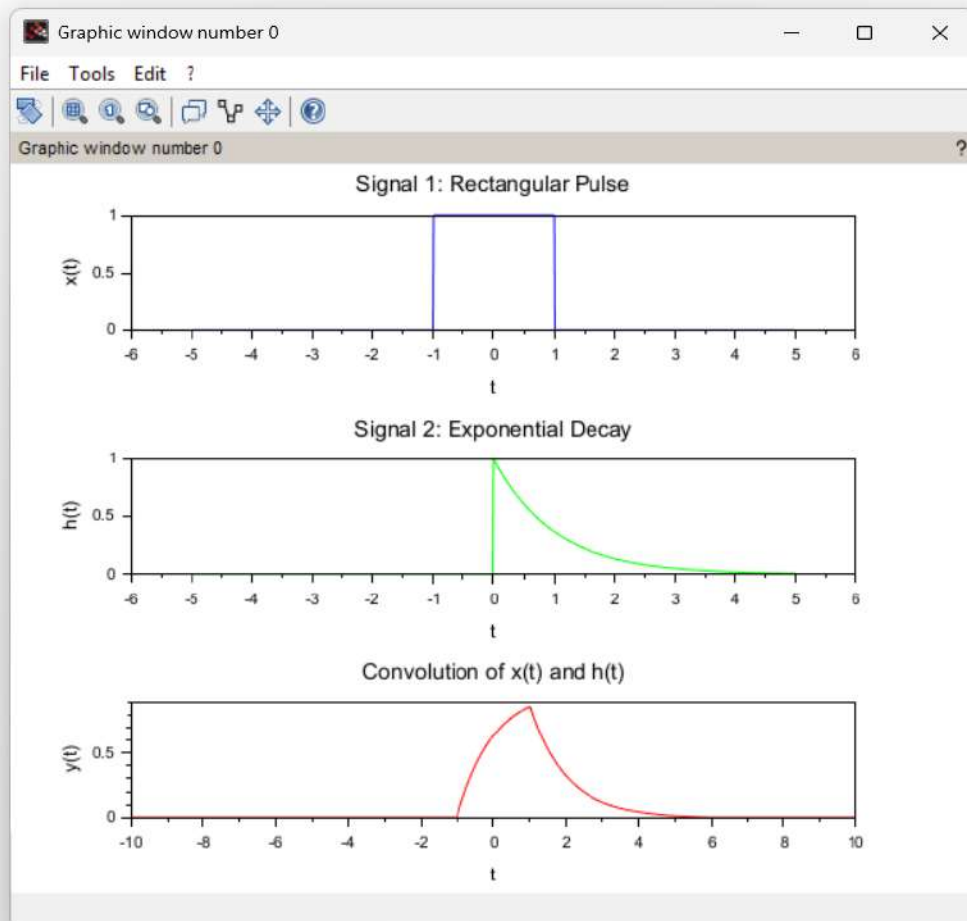


```

//Test Case: Convolution of Continuous-Time Signals.
//Define time range and sampling interval.
t_min = -5;
//Start time.
t_max = 5;
dt = 0.01;
t = t_min:dt:t_max;
//End time.
//Time step.
//Time vector.
//Define Signal 1: Rectangular pulse of width 2 centered at t = 0.
x_t = (t >= -1) .* (t <= 1);
//Define Signal 2: Exponential decay function.
h_t = exp(-t) .* (t >= 0); //Exponential decay, causal.
//Perform the convolution.
y_t = conv(x_t, h_t) * dt; //Multiply by dt to scale correctly.
//Adjust the time vector for the convolution result.
t_conv = (2 * t_min) : dt : (2 * t_max);
//Plot Signal 1.
clf; //Clear the current figure.
subplot(3,1,1);
plot(t, x_t, 'b');
xlabel('t');
ylabel('x(t)');
title('Signal 1: Rectangular Pulse');
//Plot Signal 2.
subplot(3,1,2);
plot(t, h_t, 'g');
xlabel('t');
ylabel('h(t)');
title('Signal 2: Exponential Decay');
//Plot the Convolution Result.
subplot(3,1,3);
plot(t_conv, y_t, 'r');
xlabel('t');
ylabel('y(t)');
title('Convolution of x(t) and h(t)');

```



```

1 //Test-Case:-Convolution-of-Discrete-Time-Signals-
2 //Define-the-discrete-time-signals-
3 x_n = [1, 2, 3];
4 //Signal-x[n]-
5 h_n = [1, -1];
6 //Signal-h[n]-
7 //Compute-the-convolution-
8 y_n = conv(x_n, h_n); //Convolution-of-x[n]-and-h[n]-
9 //Define-the-time-index-for-the-output-signal-
10 n_y = 0:(length(x_n) + length(h_n) - 2); //Length-of-y[n]-
11 //Plot-the-input-signals-and-their-convolution-
12 clf; //Clear-the-current-figure-
13 //Plot-x[n]-
14 subplot(3,1,1);
15 bar(0:length(x_n)-1, x_n, 'b');
16 xlabel('n');
17 ylabel('x[n]');
18 title('Signal-x[n]');
19 //Plot-h[n]-
20 subplot(3,1,2);
21 bar(0:length(h_n)-1, h_n, 'r');
22 xlabel('n');
23 ylabel('h[n]');
24 title('Signal-h[n]');
25 //Plot-y[n]- (convolution result)-
26 subplot(3,1,3);
27 bar(n_y, y_n, 'g');
28 xlabel('n');
29 ylabel('y[n]');
30 title('Convolution-of-x[n]-and-h[n]');
31

```

